

MENG DING, China

WMO: 569450

Lat: 23.567N

Lon: 99.083E

Elev: 1680

StdP: 13.83

Time Zone: 8.00 (E08)

Period: 82-89

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 46.4	(c) 48.2	(d) 42.6	(e) 42.9	(f) 66.9	(g) 44.4	(h) 45.9	(i) 62.7	(j) 11.9	(k) 77.4	(l) 10.0	(m) 76.2	(n) 0.1	(o) N/A	(p) 0.262

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 6	(b) 12.9	(c) 96.0	(d) 72.2	(e) 93.1	(f) 72.1	(g) 90.9	(h) 73.1	(i) 79.2	(j) 88.0	(k) 78.4	(l) 86.8	(m) 77.8	(n) 85.6	(o) 7.3	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.8	148.6	84.0	76.1	145.3	82.8	75.5	142.3	81.8	44.4	88.5	43.5	87.2	42.8	86.1	88.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 13.4	(b) 10.3	(c) 7.9	(d) DB	(e) 42.8	(f) 101.0	(g) 2.0	(h) 1.5	(i) 41.4	(j) 102.0	(k) 40.2	(l) 102.9	(m) 39.0	(n) 103.7	(o) 37.6	(p) 104.8	(q) 90.6
(a) 13.4	(b) 10.3	(c) 7.9	(d) WB	(e) 42.4	(f) 83.0	(g) 1.9	(h) 2.9	(i) 41.0	(j) 85.1	(k) 39.9	(l) 86.8	(m) 38.8	(n) 88.4	(o) 37.4	(p) 90.6	(q) 90.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	73.6	62.8	66.2	71.4	75.5	80.3	80.5	79.1	80.0	79.0	75.9	69.2	63.3
	DBStd	7.18	2.63	3.12	3.86	3.37	3.84	2.94	2.89	2.20	2.33	3.10	3.01	2.99
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	179	77	22	6	0	0	0	0	0	0	0	4	70
	CDD50	8628	396	454	665	766	938	915	903	929	871	804	575	412
	CDD65	3333	8	56	206	317	473	465	438	464	421	339	128	18
	CDH74	26815	152	681	2231	2795	4603	4081	3289	3604	2922	1886	485	86
	CDH80	9491	6	156	917	1255	2251	1392	910	1075	927	542	58	2
(14) Wind	WSAvg	2.0	1.3	2.0	2.4	2.6	3.0	2.4	2.3	2.3	1.9	1.4	1.1	0.9
(15) Precipitation	PrecAvg	59.8	0.6	0.7	0.6	1.7	5.0	10.9	13.3	12.4	6.7	4.9	2.6	0.6
	PrecMax	71.7	2.8	3.0	2.2	5.9	13.3	21.3	19.6	19.0	14.8	12.5	8.3	2.8
	PrecMin	49.2	0.0	0.0	0.0	0.0	1.2	4.4	7.0	5.0	2.7	1.7	0.0	0.0
	PrecStd	5.9	0.6	0.7	0.6	1.1	3.1	3.6	3.3	3.5	2.8	2.9	2.2	0.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	80.8	87.5	93.2	96.7	99.7	97.7	92.3	91.6	92.1	89.6	84.4	79.1
		MCWB	62.8	63.4	65.9	67.7	71.8	76.3	79.3	78.4	78.2	76.2	70.7	65.4
	2%	DB	78.0	84.5	90.6	93.6	97.2	91.9	89.7	89.6	89.7	87.5	81.7	76.4
		MCWB	61.5	62.7	65.4	67.9	72.7	77.2	78.6	78.2	77.6	75.1	68.4	63.9
	5%	DB	75.5	82.0	88.3	91.2	94.7	89.3	87.1	87.8	87.7	85.5	79.4	74.1
		MCWB	60.9	61.8	64.9	67.8	72.6	76.9	77.6	77.9	77.1	74.5	67.4	62.7
	10%	DB	72.7	78.7	85.6	87.6	91.2	86.9	85.0	85.8	85.3	83.3	76.9	71.3
		MCWB	59.8	61.1	64.1	67.8	72.8	76.6	76.9	77.3	76.4	73.5	66.7	61.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.6	65.2	69.1	72.8	77.6	79.7	80.1	80.0	79.9	78.0	72.5	66.9
		MCDB	76.6	82.5	86.7	85.5	88.1	89.3	90.1	88.6	88.5	86.7	79.1	76.4
	2%	WB	62.9	63.8	67.6	71.6	76.2	78.4	79.0	79.0	78.5	76.4	70.7	65.0
		MCDB	74.5	80.3	85.6	83.3	86.8	87.7	87.5	87.3	87.1	84.7	76.1	74.0
	5%	WB	61.6	62.8	66.3	70.6	75.2	77.7	78.1	78.3	77.5	75.5	69.6	63.5
		MCDB	72.9	78.2	84.0	82.7	86.0	86.1	85.6	86.0	85.5	82.9	75.7	71.6
	10%	WB	60.4	61.8	65.1	69.6	74.3	77.0	77.2	77.5	76.7	74.5	68.3	62.1
		MCDB	71.0	76.1	82.1	82.0	85.1	84.6	83.8	84.5	83.8	80.9	74.5	69.3
(35) Mean Daily Temperature Range	5% DB	MDBR	27.0	29.0	30.5	24.3	20.9	12.9	10.8	12.6	14.6	16.7	19.0	22.1
		MCDBR	28.9	33.8	34.4	31.5	26.9	17.8	15.2	15.8	18.5	19.6	22.2	23.7
		MCWBR	12.7	12.5	11.8	10.1	6.9	5.5	5.5	6.2	7.2	8.0	9.3	10.8
	5% WB	MCDBR	26.5	30.3	30.4	23.2	18.9	13.8	13.7	14.4	16.0	16.7	17.5	22.2
		MCWBR	12.1	11.9	10.9	8.6	6.2	5.1	5.6	6.0	6.7	7.0	7.7	10.4
(40) Clear-Sky Solar Irradiance	taub		0.375	0.438	0.710	0.704	0.538	0.470	0.459	0.497	0.489	0.465	0.404	0.378
	taud		2.327	2.144	1.513	1.588	2.035	2.285	2.311	2.179	2.177	2.217	2.326	2.354
	Ebn at Noon		279	267	204	210	247	262	265	256	256	256	268	274
	Edh at Noon		36	47	92	87	55	43	42	48	47	43	36	34
(44) All-Sky Solar Radiation	RadAvg		1384	1660	1719	1731	1562	1334	1220	1347	1362	1296	1334	1231
	RadStd		97	119	124	132	162	125	85	105	95	137	100	94

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	+0.45	N/A	N/A	N/A	+0.76	N/A	N/A	N/A	N/A	+166	+99

Nomenclature: See separate page